

Section 3.5: Optical Storage Media

In this section you will see the 8 different types of optical storage media.

- CD-ROM
- DVD-ROM
- CD-R
- DVD-R
- CD-RW
- DVD-RW
- Blu-Ray Disc
- DVD-RAM

For each we will discuss **features, uses, advantages/disadvantages**.

Key Concepts of this section:

- # Know the **different types** of optical storage media.
- # Understand the **uses** of each optical media.
- # Be able to **discuss advantages and disadvantages** of each optical media.

Optical Storage Media (an overview)

Definition: "All the mediums that use a light source to read/write data onto a disc."

Overview of Optical Storage Media:

- # Optical storage media stores data on a **disc** as a number of **data dots** that can be **read using light** (usually a laser beam).

Each dot represents **1's and 0's** (Bits of information).

- # Data is read by **shining the laser beam** onto the **surface of the disc**.

If the light hits one of the dots it is **reflected back differently** than it would be if there was no dot. This difference is **read as data** by the computer.

- # Some Optical Disks (such as CD-R's) allow you to **write data** to the disc as well as read it.

This works by using the laser beam to **'burn'** dots onto the surface of the disk (creating the data) and then reading them back again. This process is known as **'burning'**.

- # In order to read/write data on a disc you need a **'disc drive'**.

For example: CD Drive, DVD Drive, Blu-Ray Drive

The drive contains the following:

- **A mount for the disc**
- **A laser** (to read/write the data bumps)
- **A motor** (to spin the disc across the laser)

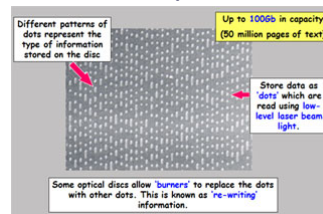
Animation showing how Disc-Drives work.



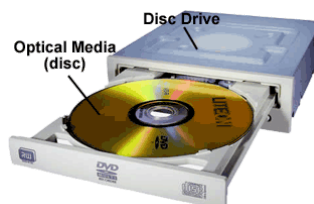
Key Words:

Data Dots, Disc, Laser, Burn, Disc Drive

Examples:



Data is stored on optical media in the form of tiny bumps known as 'data dots'. (Click to zoom)



A DVD disc drive is used to read and add data to DVD optical disc.

NOTE:

All the examples of optical storage media on this page use **direct data access**.

Examples of Optical Media

- # The table below shows you some examples of optical storage media. Click each of the media to move to more information:



Key Words:

CD-ROM, DVD-ROM, CD-R, DVD-R, CD-RWR, DVD-RW, DVD-RAM, Blu-Ray Disc

CD-ROM



(Direct Access)

DVD-ROM



(Direct Access)

CD-R



(Direct Access)

DVD-R



(Direct Access)

CD-RW



DVD-RW



DVD-RAM



Blu-ray Disc



REMEMBER!

Optical storage media stores data on tiny bumps called data dots.

NOTE:

Click the images to the left to move to more information about each one.

(Direct Access)

(Direct Access)

(Direct Access)

(Direct Access)

Breakdown of each output device:

The following sections will help summarise each of the optical storage media. For each one you will see:

- **Information about the media**
- **Uses of the media**
- **Advantages of the media**
- **Disadvantages of the media**

1 CD-ROM and DVD-ROM

Information:

ROM stands for 'read only memory'.

This means that the data **cannot be written over** (added to) and can **only be read**.

CD-ROM's and DVD-ROM's are bought with **pre-loaded data** already on them.

Usually pre-loaded with **software, music, movies and games**.

Whatever is already loaded onto the ROM **cannot be removed or changed**.

Remember: This is because ROM discs are 'read only'.

So if you purchase a music CD, you can listen to the tracks on the disc but you can't add more songs of your own.

CD-ROMs have a maximum **capacity** of about **800MB** (about 400,000 pages of text)

DVD-ROMs have a maximum **capacity** of about **4.7GB** (about 2.4 million pages of text)

Uses of CD-ROMs and DVD-ROMs:

CD-ROMs are used by manufacturers to store **smaller files** (up to 800MB) such as:

- **Music tracks**
- **Software** (such as word processors, spreadsheets etc)
- **Audibooks**
- **etc**

DVD-ROMs have much **larger storage capacities** (up to 4.7 GB) than CD-ROMs and are used to store bigger files such as:

- **Movies**
- **Modern games** (like grand theft auto, call of duty etc)
- **Multimedia encyclopedias**
- **etc**

Examples:

CD-ROMs store up to 800MB and were used to distribute pre-loaded software, music etc.



DVD-ROMs are used to distribute larger files such as movies.

Advantages/Disadvantages of CD-ROMs and DVD-ROMs:

Advantages	Disadvantages
Hold far more data than Floppy Discs.	Hold far less data than hard disk drives or memory sticks.
Less expensive than Hard Disc Drives.	Very slow data transfer and access rates when compared to hard disk drives. (Reading back from the discs is slow)
Data stored on them is ' read only ' which means you cannot accidentally erase the files.	You cannot add your own files to the discs because they are both read only.
Very portable as they are small and light.	

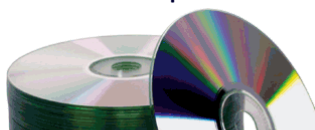
2 CD-R and DVD-R

Information:

The letter '**R**' means that the disc is **recordable once** only.

Once the disc **has data recorded** on it becomes a **CD/DVD ROM** (read only).

Data can be added to the disc (once) but **NOT erased** when it is on there.

Examples:

- # Data is '**burnt**' onto the discs using a **disc burner**.

Note: A disc burner works in the same way as a regular disc drive but it uses a **more powerful laser** to burn data onto a CD/DVD-R disc.

A disc burner **can also read the data** held on the disc using a **less powerful laser** that is not strong enough to burn.

- # Thin layer of **metallic dye** is used to record the data onto.

When CD-R's and DVD-R's are burnt, the laser makes **permanent marks** (dots of data which represent 1's and 0's) onto the metallic dye.

These marks **cannot be erased** (which is why what you add to the disc is permanent).

DVD-R's use a dye that allows for **more dots** to be burnt onto the disc's surface.

This is why DVD-R's can hold more data than CD-R's

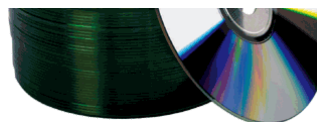
A disc has data 'burnt' onto its surface using a more powerful laser.

- # **CD-R** have a maximum **capacity** of about **700MB** (about 80 minutes of audio)

DVD-R have a maximum **capacity** of about **4.7GB** (over 2 hours of video)

Note: DVD-R's are also available in '**dual-layer**'.

This means they have **two layers** of recordable surfaces which means they can store more data (about 8.5GB) than a regular DVD-R.



CD-R's can have about 700MB of data burnt onto a special metallic dye.



DVD-R's use a different type of metallic dye that allows for more data to be held on the disc.



A DVD burner adds data onto a disc and reads it back off.



Dual Layer DVD's allow for more data to be held on the disc.

Uses of CD-R and DVD-R:

- # CD-R's used to create **personalised playlists** of music.
- # DVD-R's used to store **home recordings of video** shot on digital cameras.
- # They can be used to **backup important data** so that there are copies in case the originals get erased or deleted.
- # They can be used to **transfer data** from one computer to another.

Advantages/Disadvantages of CD-R and DVD-R:

Advantages	Disadvantages
Once burned, data is impossible to accidentally delete because the disc becomes read only.	Only recordable once. This means updating the disc is impossible .
Cheaper than RW discs and Hard Disk Drives .	If an error occurs during burning the disc is damaged and must be thrown away (wasted)
Easy to transport information from one computer to another (Can take the disc out of one machine and use in another)	Very slow data transfer rate. (Burning data onto a disc can take up to 15 minutes) Not all CD/DVD players can read CD-R and DVD-R discs



Data is burnt onto CD-R/DVD-R discs using a laser.

Once the data is on the disc it cannot be removed.

3 CD-RW and DVD-RW

Information:

- # The letters '**RW**' stand for **Re-Writeable**. This means that the disc can be **recorded on over and over** again.
- # Unlike CD/DVD-R's these discs **DO NOT become ROMS** (not read only) once burned.
- # **RW drives** are used to burn data onto dye on the RW disc (and also read from it).
- # The dye used to record data is called '**phase changing dye**' and it allows the bumps of data to be '**undone**'.

The **bumps of data** are really just **flattened again** which effectively **erases** any stored data.

- # Data can be **added to the disc** and can also be **erased**.

CD-RW's and DVD-RW's can be **reused** around **1000 times**.

- # **CD-RW** have a maximum **capacity** of about **700MB**

Examples:



CD-RW can store up to 700MB of data and be reused over and over.

DVD-RW have a maximum **capacity** of about **4.7GB**

Uses of CD-RW and DVD-RW:

- # CD-RW's are used to create **personalised playlists** of **music** and then **add/remove songs** as you wish.
- # DVD-RW's are used to **record television programmes** and can be **recorded over** many times.

DVD-RW's are also used to **record video** content of **closed circuit television (CCTV)** to allow security to keep an eye on businesses and what is happening on the streets

- # They can both also be used to make **backups** of computer files.



DVD-RW can store up to 4.7GB of data and be reused over and over.

Advantages/Disadvantages of CD-RW and DVD-RW:

Advantages	Disadvantages
Can be re-used many times .	More expensive to buy than CD/DVD-R discs
Data stored on the discs can be updated .	It is possible to accidentally overwrite data (since RW discs can be updated) .
Not as wasteful as the -R format. (Even if burning fails, the disk can still be recorded on later and not thrown away)	Data transfer to the discs is very slow compared to a hard disk drive. (Depending on the speed of your disc burner it can take up to 15 minutes to burn an entire DVD-RW)

4 DVD-RAM (DVD-Random Access Memory)

Information:

- # With DVD-RAM **writing and reading** of data can happen **at the same time**.

This means that you could watch a programme **at the same time** that another one is being recorded - (Read and write at the same time)

- # They use a **similar Phase Changing Recording Dye** to **CD/DVD-RW's** which allows DVD-RAM discs to be recorded over many times.

Note: DVD-RAM is **much faster** and **easier** than CD/DVD-RW at overwriting data.

- # DVD-RAM discs can be **rewritten over many many times** (over 100,000).
- # Data can be **reliably stored** on DVD-RAM for many years due to their high quality.
- # DVD-RAM has a **capacity** of **4.7GB** (or 9.4GB with with double sided discs).

Uses of DVD-RAM:

- # Because DVD-RAM discs last so long they are used in **video and data archiving** (Safe store for important files and records).
- # Used in **DVD recorders** to **record TV programmes** (on digital and satellite) to allow simultaneous recording and playback.
- # Used in **camcorders** to store **video** (Reliably and for many years).

Examples:



DVD-RAM is a very high quality method of storing data. Discs can last for years and years.



DVD-RAM is used in DVD recorders to record TV programmes.

Advantages/Disadvantages of DVD -RAM:

Advantages	Disadvantages
Long life - last at least 30 years.	Expensive compared to DVD and CD recordable discs.
Can be written over 100,000 times (RW Disks only allow 1,000 re-writes).	Don't work in as many devices as the -R or -RW discs.
Very fast access to stored files .	
Offer very large storage capacity compared to CD's (Up to 9.4Gb with double sided discs).	
Can read data at the same time it is being written .	

5 Blu-ray Disc

Information:

- # Work in a similar way to DVD ROMS but the **laser used** to **read the data** is **blue** rather than **red** (Red lasers used to read the other disc types).

This blue laser colour is why the name '**Blu-ray**' was used

- # Blue lasers are capable of **reading data dots** that are **positioned closer together** on the disk surface.

As a result, **more data dots** (1's and 0's) can be **stored and read**.

Remember: More data dots = **higher capacity**. (up to 50 million pages of text)

- # Blu-ray-RW discs can be **rewritten** in much the same way as RW disks.
- # Blu-ray's have **capacities** of **25GB, 50Gb** and **100GB** (depending on the disc).

Uses of Blu-ray Discs:

- # Used to store **high definition video** (Like high quality movies).

One 25GB Blu-ray disc can store **2 hours** of **HDTV**.

- # Used in some **home video consoles** (Like Playstation 3).
- # Used to **back up** hard disk drives in PC's.

Advantages/Disadvantages of Blu-ray:

Advantages	Disadvantages
Huge storage capacity. Perfect for high definition movies .	Very expensive compared to other optical media.
Data can be read/transferred very fast when compared to other optical media.	Only work in Blu-Ray drives/players which are expensive. (Not everyone has these drives which limits the use of the disks)

Examples:

A Blu-ray disc and drive.



Blu-ray discs have huge storage capacities.



Blu-ray discs are read with a blue laser. This laser can read data dots that are closer together..



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